

Between Narrator Analysis in Frankenstein

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Introduction

- Frankenstein is told through multiple narrators
- Goal: determine if Mary Shelley created measurably distinct voicing between each narrators
- Data source: Frankenstein_1831.txt

Dividing by narrator

```
file = open("frankenstein_1831.txt", "r", encoding="utf-8")
raw_text = file.read()
file.close()

#removes gutenbergl licence
end_marker = "*** END OF THE PROJECT GUTENBERG EBOOK"
raw_text = raw_text.split(end_marker)[0]

#finds narrator boundaries
letter1 = raw_text.find("\nLetter 1\n")
chapter1 = raw_text.find("\nChapter 1\n")
chapter11 = raw_text.find("\nChapter 11\n")
chapter17 = raw_text.find("\nChapter 17\n")
continuation = raw_text.find("Walton, _in continuation._")

#assigns each section to each narrator
walton_text = raw_text[letter1:chapter1] + raw_text[continuation:]
victor_text = raw_text[chapter1:chapter11] + raw_text[chapter17:continuation]
creature_text = raw_text[chapter11:chapter17]
```

Finding distance between narrators

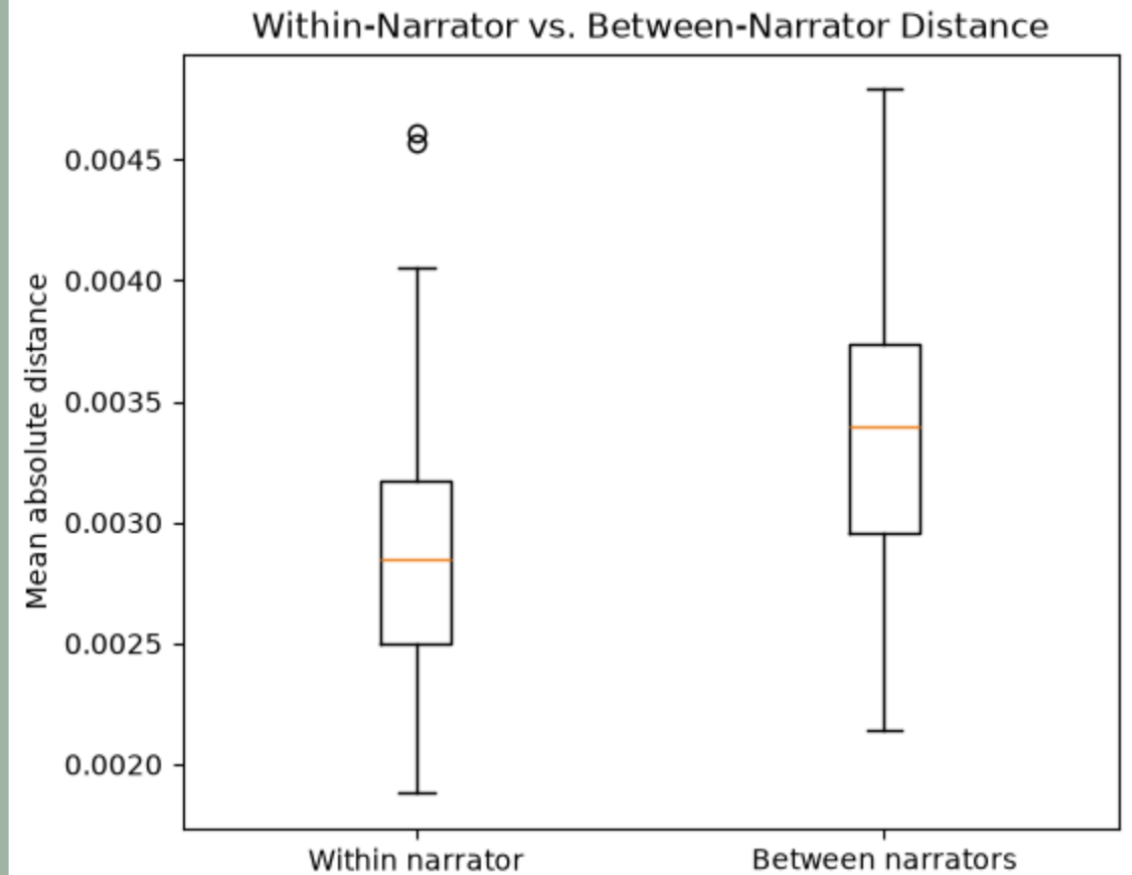
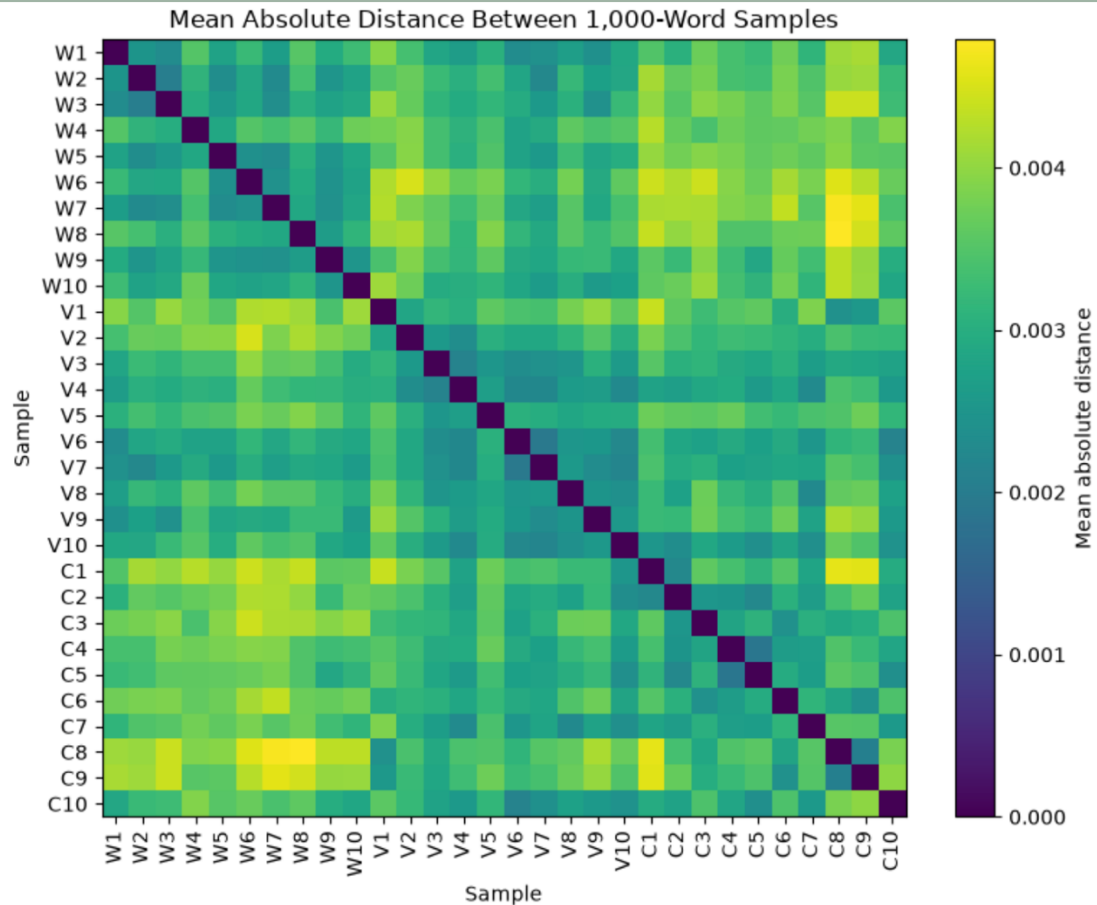
	sample	narrator	the	and	of	i	to	my	a	in	...	there	even	its	never	up	made	while	said	towards
0	W1	Walton	0.058	0.035	0.035	0.035	0.029	0.038	0.024	0.021	...	0.004	0.001	0.003	0.002	0.001	0.002	0.000	0.000	0.001
1	W2	Walton	0.041	0.035	0.037	0.041	0.032	0.022	0.027	0.020	...	0.003	0.002	0.001	0.004	0.000	0.001	0.000	0.000	0.001
2	W3	Walton	0.042	0.026	0.026	0.035	0.034	0.028	0.026	0.014	...	0.002	0.001	0.001	0.002	0.000	0.000	0.001	0.000	0.003
3	W4	Walton	0.060	0.027	0.029	0.021	0.030	0.005	0.027	0.019	...	0.002	0.001	0.000	0.003	0.005	0.000	0.000	0.003	0.003
4	W5	Walton	0.043	0.034	0.031	0.027	0.029	0.028	0.022	0.010	...	0.000	0.002	0.000	0.001	0.002	0.001	0.000	0.002	0.000

```
def mean_absolute_distance(a, b):  
    #find absolute difference  
    differences = np.abs(a - b)  
    #find average difference  
    distance = np.mean(differences)  
    return distance
```

```
#empty distance matrix  
distance_matrix = np.zeros((len(frequency_matrix), len(frequency_matrix)))  
  
#compares the frequencies in each chunk to all the others  
for i in range(len(frequency_matrix)):  
    for j in range(len(frequency_matrix)):  
        distance_matrix[i, j] = mean_absolute_distance(frequency_matrix[i], frequency_matrix[j])  
  
comparisons = []  
  
#checks if the comparisons are with other narrators or the same narrator  
for i in range(len(frequency_matrix)):  
    for j in range(i + 1, len(frequency_matrix)):  
        if narrators[i] == narrators[j]:  
            comparison_type = "Within narrator"  
        else:  
            comparison_type = "Between narrators"  
  
    #adds the type into the comparisons list  
    comparisons.append({  
        "sample_1": labels[i],  
        "sample_2": labels[j],  
        "type": comparison_type,  
        "distance": distance_matrix[i, j]  
    })
```

Between narrators	0.003386
Within narrator	0.002892

Visualizing the data



Conclusion and Further Investigation

- There are measurably different word-frequency patterns
- The difference is very small (0.0005)
- Different distance algorithms and different methods of splitting the data could be used to potentially get further insights on the voicing patterns.

The End